
Modality-Driven Search with Holistic Trace Judging for ARC-AGI-2

72.9% verified result, beating frontier models by ~20%

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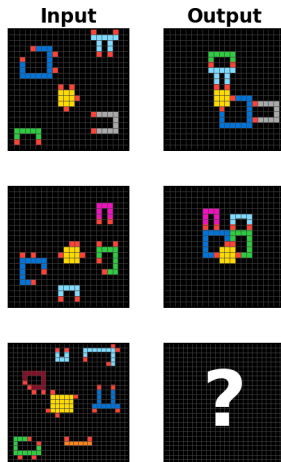
ARC-AGI-2 — a brief history, and two example problems

- **2019** — ARC-AGI-1 introduced (Chollet); trivial for humans, impossible for AI at the time
- **2020–2023** — Program synthesis era; top scores reach $\sim 20\%$ on ARC-AGI-1
- **Dec 2024** — o3 saturates ARC-AGI-1 ($\sim 76\%$ at \$200/task), but only $\sim 3\%$ on ARC-AGI-2
- **2025** — Focus shifts to ARC-AGI-2; top scores climb to $\sim 50\%$ (GPT-5.2, Opus 4.5)
- **Dec 2025** — **This work: 72.9% on ARC-AGI-2**, first verified $>70\%$
- **2026** — Frontier models leapfrog to $\sim 80\%+$ (GPT-5.5, Gemini 3.1 Deep Think, Opus 4.7); ARC-AGI-3 launches ($<1\%$)

ARC-AGI Example 1



ARC-AGI Example 2



What is ARC-AGI-2? — format and evaluation

What the model sees (JSON):

```
{
  "train": [
    {"input":  [[0,1,0],[1,0,1],[0,1,0]],
      "output": [[1,0,1],[0,1,0],[1,0,1]]},
    {"input":  [...], "output": [...]},
    ...
  ],
  "test": [
    {"input":  [[1,0,1],[0,1,0],[1,0,1]]}
  ]
}
```

What the model returns:

```
[[0,1,0],[1,0,1],[0,1,0]]
```

- **pass@2**: the model is allowed *two guesses* per test input — it scores if either matches the ground truth exactly
- **Public eval** for development; the **private eval** is the official scoring — run end-to-end by the ARC Prize foundation on tasks the developer never sees
- **Cost per task** (USD) is graded alongside accuracy — efficiency is a first-class metric

The headline result

System	ARC-AGI-2	Cost/Task
Human Panel	100.0%	\$17.00
This work (private)	72.9%	\$38.99
This work (public eval)	76.1%	\$19.69
GPT-5.2 Pro (High)	54.2%	\$15.72
Opus 4.5 (Thinking, 64K)	37.6%	\$2.40
Gemini 3 Pro	31.1%	\$0.81

+18.7 points over the strongest standalone frontier model

Two problems: break groupthink, then recognize the outlier

- LLMs produce fluent, internally coherent traces — and are still **confidently wrong**
- On hard tasks, models **cluster around the same wrong interpretation**
- $\Rightarrow$ majority voting / self-consistency *amplifies* the error
- **(1) Break the groupthink** — diversify so some candidate reaches the correct answer
- **(2) Recognize the rare-correct** — pick it out when most others disagree

Roadmap: (1) next two slides · (2) after that · then both wired together · then on a real task

(1) Break the groupthink — modalities as search operators

TEXT

Model reads the grid as numbers / strings

IMAGE

Model sees a rendered PNG of the grid

CODE

Model writes Python to transform the grid

- Same task, three representations → three different hypothesis distributions
- Each modality solves tasks the others miss
- Generate **independently** — don't pool prompts, don't share scaffolding
- Diversity at the *representation* level, not just temperature

(1) Break the groupthink — candidate generation in practice

- **3 models:** GPT-5.2 (x-high), Gemini 3 Preview (high), Claude Opus 4.5 (long-context)
- **3 modality families:** Text, Image, Code
- **29 candidates per task** across families × configurations
- **Deliberately minimal prompts** in every lane (*see slide 10*)
- Code candidates use a sandbox REPL for iterative debugging

(2) Recognize the rare-correct — holistic judging vs. majority voting

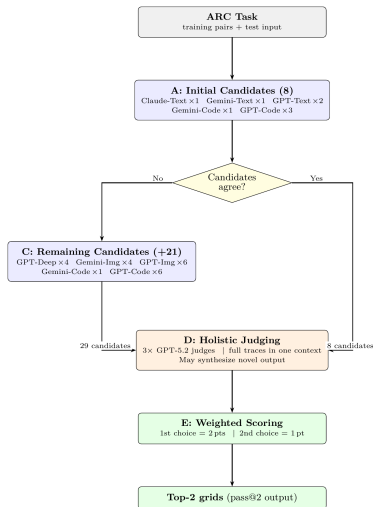
- Concatenate all 29 reasoning traces into one long-context judge prompt
- Each judge picks top-2 candidates *and* explains why other clusters are wrong
- Weighted vote across 3 judges → pass@2 guesses

Net uplift vs. majority-vote baseline: +7 instances

All 7 are minority recoveries — the correct answer wasn't the most common candidate.

Plus +1 **synthesis** instance (21897d95:2): zero correct candidates — judge recombined partial insights into a novel correct output.

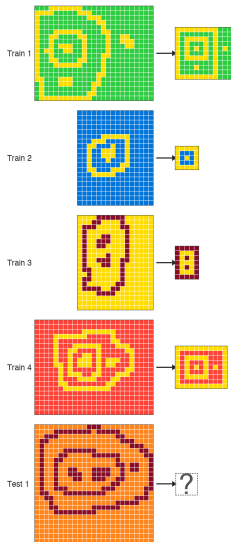
(1) + (2) wired together — pipeline overview



Generate broadly → judge holistically → vote weighted

- Up to 29 candidates across Text / Image / Code, 3 frontier models
- 3 parallel judges read *all* traces in one long-context prompt
- Adaptive early-stop: 37/167 stopped after first probe; **36 correct (97%)**

(1) + (2) on a real task — 28 of 29 candidates wrong, judges picked the right one



Public eval task **2d0172a1:1** — candidate pool $n = 29$

Modality family	Correct	Wrong
Text (8 candidates)	0	8
Image (10 candidates)	1	9
Code (11 candidates)	0	11
Total	1	28

Submission verdict: SOLVED — holistic judges selected the lone correct candidate

- One image candidate produced the correct nested-frame output; 28 of 29 were wrong
- Majority voting would collapse onto the wrong cluster
- **Our judges recovered the minority-correct candidate**

(2) Recognize the rare-correct — what the judges actually read (2d0172a1:1)

This task produced **31,208 lines** of reasoning traces / logs.

Winning candidate

GPT-5.2 x-high, image modality

```
// logs/2026-01-04_22-09-54_2d0172a1_1_step_5.json
{
  "image": {
    ...
    "gpt-5.2-xhigh_13_step_5_image...": {
      "actual_model": "gpt-5.2-xhigh",
      "input_tokens": 21188,
      "output_tokens": 59323,
      "Full raw LLM response":
      "### What the examples show

In every example, the foreground pixels form
several separate components of two types:
1. Frames (loops): ... every such component
becomes a perfect axis-aligned
rectangular border (thickness 1).
2. Markers (solid small components): ...
each marker is reduced to a
single foreground cell.
...
### Applying to the test input
- Background color = 7, foreground = 9.
- One outer loop frame, one inner loop
frame inside it, ... two markers
inside, side-by-side  $\Rightarrow$  horizontal.
... outer size 9 rows  $\times$  13 columns",
      "Extracted grid": [...]
    }, ...
  }, ...
}
```

Judge's selection rationale

GPT-5.2 x-high $\times$ 3; all 3 picked correctly

```
// logs/2026-01-04_22-09-54_2d0172a1_1_step_finish.json
{
  ...
  "selection_details": {
    "judges": {
      "duo_pick_council": [
        ..., {
          "model": "gpt-5.2-xhigh",
          "response":
          "Across the four solved examples, the
consistent mechanic is topological
normalization:
- Each loop becomes a perfect 1-cell-
thick rectangular frame.
- Each solid component becomes a
single foreground cell (a dot).
...
The two that best match the examples and
the apparent test structure are:
- Solution 10: outer frame contains an
inner frame plus an extra dot to the
right (matching the extra small
disconnected 9-component at cols
~20--21 in the test input) ...
- Solution 6/7: the same, but without
the extra right-side dot.
### Candidate 1 (most likely): Solution 10",
          "picked_grids": [...]
        }, ...
      ]
    }
  },
  ...
}
```

(1) + (2) across 130 tasks — modality contributions and judges' verdicts



How to read (one row = one of **130 instances** with full 29-candidate coverage):

- **Judges column** (leftmost): the system's final pass@2 verdict — did the holistic judges submit a correct answer?
- Other columns grouped by modality family — **Text** · **Image** · **Code** — each sub-column = one model/config candidate
- **green** = correct · **red** = wrong
- Rows sorted by difficulty: hardest tasks top, easiest bottom

Three bands tell the story:

- **Top:** all-red rows — generation failures (no modality solved them)
- **Middle:** *scattered green cells in mostly-red rows* $\Rightarrow$ different modalities catch different problems
- **Bottom:** mostly green — the easy tasks (many modalities solve)

Edge case — Judges green, all candidates red: a *judge-synthesis* solve.

Happens on 21897d95 : 2: zero candidates correct, but a judge **recombined partial insights** across failed candidates into a novel correct output.

What the numbers say ($n=130$):

- Exclusive solves: **2 Text-only** · **6 Image-only** · **7 Code-only**

Negative results worth knowing

- 1. Prescriptive prompting degrades performance.** \ Templates, CoT scaffolding, domain heuristics — all hurt on the hardest tasks. Mechanism: a *compliance tax on reasoning* (budget spent following the template instead of solving the problem).
- 2. Iterative refinement reduces diversity.** \ Refining candidates against each other collapses them onto the same wrong cluster — the exact failure mode we're trying to escape. Same effect from hint-then-solve pipelines.
- 3. Pipeline decomposition (objects → transformations → solver) hurts.** \ Brittle handoffs between stages. Both verbose and terse handovers regress toward the mean rather than expand the hypothesis space.
- 4. Strict JSON outputs underperform CSV.** \ Forcing API-level response-format schemas consistently lost to CSV + robust regex parsing. Strict schemas appear to constrain reasoning quality, not just output shape.
- 5. Pixel-perfect image renderings hurt.** \ *Slightly distorted* renders beat pixel-perfect ones for image-modality candidates. Clean renders push the model into cell-by-cell numerical reasoning; intentional imprecision forces engagement with shapes, symmetries, and spatial relationships — which is the point of image prompting.

⇒ **Final system: deliberately minimal prompts, independent generation, no decomposition, CSV I/O, deliberately fuzzy image renders.**

Takeaways

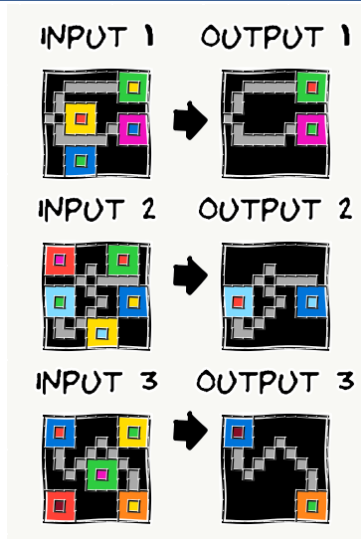
(1) Generate broadly — different modalities (text, image, code) and different frontier models, generated *independently*. The hypothesis space matters more than the model.

(2) Judge holistically — a long-context judge reads the *full reasoning*, not just final answers. Majority voting discards the rare-correct; trace-aware judging recovers it (and occasionally synthesizes a new one).

Where this generalizes: any hard problem where the *reasoning* signals correctness — even when the answer itself can't be verified directly. As in ARC: the judges never see the test output; they assess the **credibility of each trace** against the training pairs. Math contests, code review, legal arguments, scientific hypotheses — wherever a careful reader can tell sound reasoning from rationalization.

Code: github.com/beetree/ARC-AGI · Paper QR on title slide

End-to-end walkthrough: d35bdbdc



From 29 candidates → 2 guesses

Backup: cost breakdown

- **Public-eval cost is the representative number: \$19.69/task**
- Private \$38.99/task inflated by GPT-5.2 API instability (84% failure rate, 2,216/14,106 attempts succeeded)
- Tool-integrated code generation is the largest line item
- ZDR mode used in private verification disables tool calls → both lower accuracy and altered cost profile
- ARC-AGI-1 on the same system: 94.5% at \$11.40/task — easier tasks early-exit cheaply

Backup: failure decomposition (167 instances)

- **128 solved (76.6%)**
- 21 generation failures (no correct candidate in pool)
- 1 early-stop failure (dbff022c:1 — extreme groupthink on a legend ambiguity)
- 17 **selection** failures (correct candidate exists, judge missed it)
- 1 synthesis success (21897d95:2)

Most future headroom is in selection, not generation.

Backup: why three judges?

- Single judge is biased toward its own prior
- 3 parallel judges + weighted scoring (2 pts first, 1 pt second) hedges *within* the judging step
- Pass@2 format lets the final output also hedge across two interpretations when judges disagree